Chemical Equilibrium

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Types of Chemical Reactions

On The Basis of Extent of Reaction

(I) Irreversible Reaction: Reaction goes to completion or almost completion or reaction which proceeds in single direction.

Example:

Combustion Reactions:

$C(s) + O_2(g) \longrightarrow CO_2(g)$

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$

Gaseous reactions in open vessel:

$NH_4Cl(s) \longrightarrow NH_3(g) + HCl(g)$

$NH_4HS(s) \longrightarrow NH_3(g) + H_2S(g)$

Precipitation Reactions:

$$AgNO_3(aq.) + NaCl(aq.) \longrightarrow \underset{\text{Precipitate}}{AgCl(s)\downarrow} + NaNO_3(aq.)$$

Acid-Base Reactions

$$\underset{\text{acid}}{HCl(aq.)} + \underset{\text{base}}{NaOH(aq.)} \longrightarrow \underset{\text{salt}}{NaCl(aq.)} + H_2O(l)$$

$$HNO_3(aq.) + KOH(aq.) \longrightarrow KNO_3(aq.) + H_2O(l)$$

Redox Reaction

$Zn(s) + Cu^{2+}(aq.) \longrightarrow Zn^{2+}(aq.) + Cu(s)$

Represented by: Reactants $\longrightarrow$ Products

In general performed in open vessel.

Note: Irreversible reaction have 100% extent of reaction.

(II) Reversible Reactions:

Reaction which proceeds in both direction simultaneously under similar conditions.

Represented by: Reactants $\underset{\text{Backward}}{\overset{\text{Forward}}{\rightleftharpoons}}$ Products.

Performed in closed vessel.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$$2HI(g) \underset{\text{backward reaction}}{\overset{\text{forward reaction}}{\rightleftharpoons}} H_2(g) + I_2(g)$$

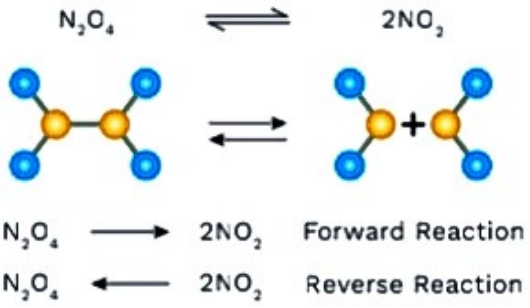
Open vessel:

$$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)\uparrow$$

Closed vessel: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

A reversible reaction never goes to complete conversion (i.e. extent of reaction ≠ 100%, but < 100%) but always achieves a state of equilibrium.

Reversible Reaction



MATERIAL EQUILIBRIUM

Chemical Equilibrium

Chemical changes involve

$$MgCO_3(s) \rightleftharpoons MgO(s) + CO_2(g)$$

$$H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$$

Phase or Physical Equilibrium

Only Phase change involves

$$X(s) \rightleftharpoons X(l)$$

$$X(l) \rightleftharpoons X(g)$$

$$H_2O(s) \rightleftharpoons H_2O(g)$$

Basic Concepts of Chemical Equilibrium

$$aA(g) + bB(g) \underset{\text{Backward}}{\overset{\text{Forward}}{\rightleftharpoons}} cC(g)$$

According to rate law:

Rate of forward direction reaction (r_f) $\propto [A]^a[B]^b$

Rate of backward direction reaction (r_b) $\propto [C]^c$

Initially r_f is maximum which decrease with time, as [A] and [B] decreases with time.

Initially r_b is zero which increases with time as [C] will increase with time.

After certain duration of time both rates become equal and then remains same until no further change is done. At this stage of time the reaction is said to be in state of equilibrium.

At equilibrium:

Rate of forward reaction (r_f) = Rate of backward reaction (r_b)

Measurable properties of system at equilibrium remains constant.
(Moles/Temp/Pressure)

	$R \rightleftharpoons 2P$	
$t = 0$	a	–
$t = t_{eq.}$	a – x	2x
$t > t_{eq.}$	a – x	2x

where a and x may be moles, molarity or partial pressure.

Closed vessel is always used.

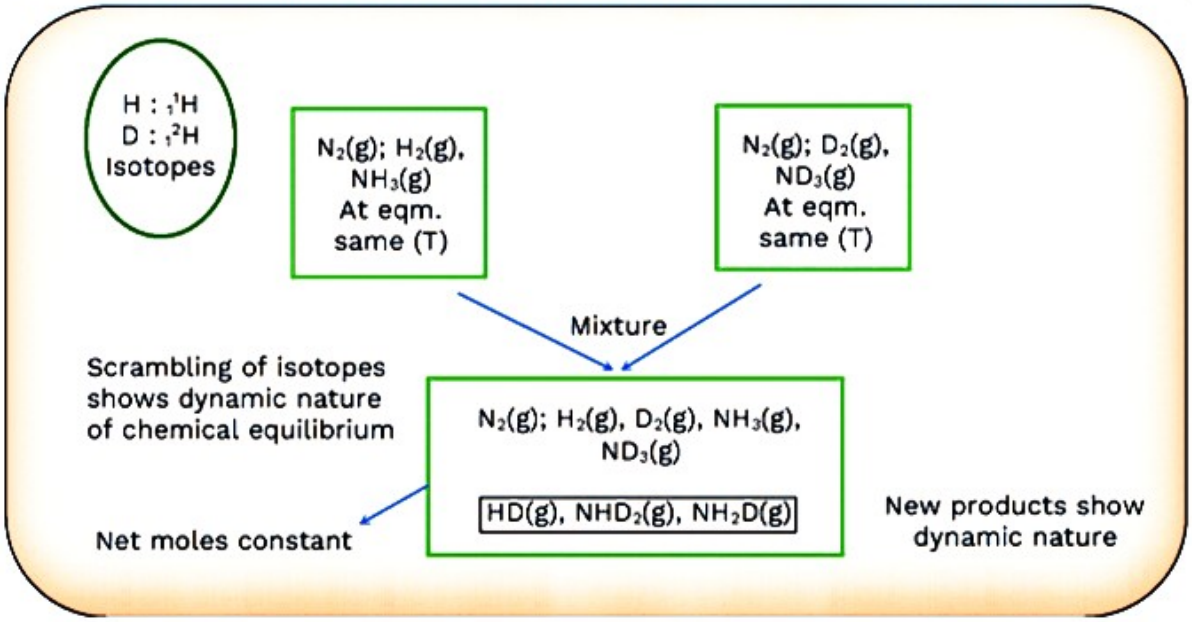
Chemical reaction is dynamics in nature.

We can achieve equilibrium from both sides:

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

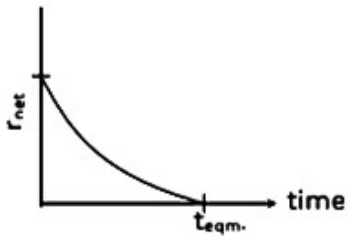
Chemical equilibrium has dynamic nature which may be observed by scrambling of isotopes.



Overall rate of a reversible reaction:

Overall rate (r_{net}) = $r_f - r_b$

$$R \underset{r_b}{\overset{r_f}{\rightleftharpoons}} P$$



At t = 0, r_f is maximum, $r_b = 0 \Rightarrow r_{net}$ is maximum

Between $0 < t < t_{eqm}$, $r_f \downarrow$, $r_b \uparrow \Rightarrow r_{net} \downarrow$

For $t \geq t_{eqm.}$, $r_f = r_b \Rightarrow r_{net} = 0$

Catalyst increases rate of reaction in forward direction as well as in backward direction with same extent, addition of catalyst or presence of catalyst never change the equilibrium composition. Catalyst only alters time in which equilibrium can be achieved.

If a reactions takes place at non-standard states then at equilibrium

$$\boxed{(\Delta_r G)_{T,P(\text{at same temperature})} = 0 \quad \text{and} \quad \Delta S_{universe} = 0}$$

$\Delta_r G$ = Change in Gibbs free energy if reaction takes place at non-standard states.

If reaction takes place at standard state then at equilibrium

$$\boxed{\Delta_r G^\circ = 0}$$

at constant temperature

ΔrG° = Change in Gibb's energy if reaction at standard states.

Law of Chemical Equilibrium (or Law of Mass Action)

Given by Guldberg and Waage

$$aA(g) + bB(g) \underset{B.D.}{\overset{F.D.}{\rightleftharpoons}} cC(g) + dD(g)$$

At constant temperature,

Rate of reaction in forward direction $r_f \propto [A]^a [B]^b$

$$\boxed{r_f = k_f [A]^a [B]^b}$$

Rate of reaction in backward direction $r_b \propto [C]^c [D]^d$

$$\boxed{r_b = k_b [C]^c [D]^d}$$

[ ]: concentration in mol/L

k_f and k_b : Rate constant of forward and backward direction reaction respectively (Changes with temperature only).

At Equilibrium:

$$\boxed{(r_f)_{equilibrium} = (r_b)_{equilibrium}}$$

$$k_f [A]_{eq}^a [B]_{eq}^b = k_b [C]_{eq}^c [D]_{eq}^d$$

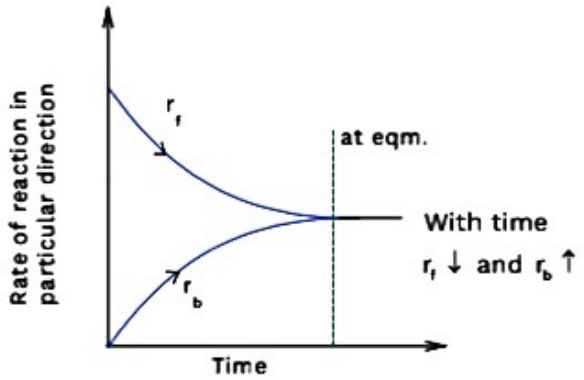
$$\frac{k_f}{k_b} = \boxed{K_{eq} = K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}}$$

$K_{eqm.}$ = Equilibrium constant.

K_c = Equilibrium constant in terms of concentration.

Note: In above expression all concentrations are at equilibrium time.

Graph:



In Kinetics

$X(g) + Y(g) \rightleftharpoons Z(g)$

Net rate of reaction =

Rate of reaction in forward direction – Rate of reaction in backward direction.

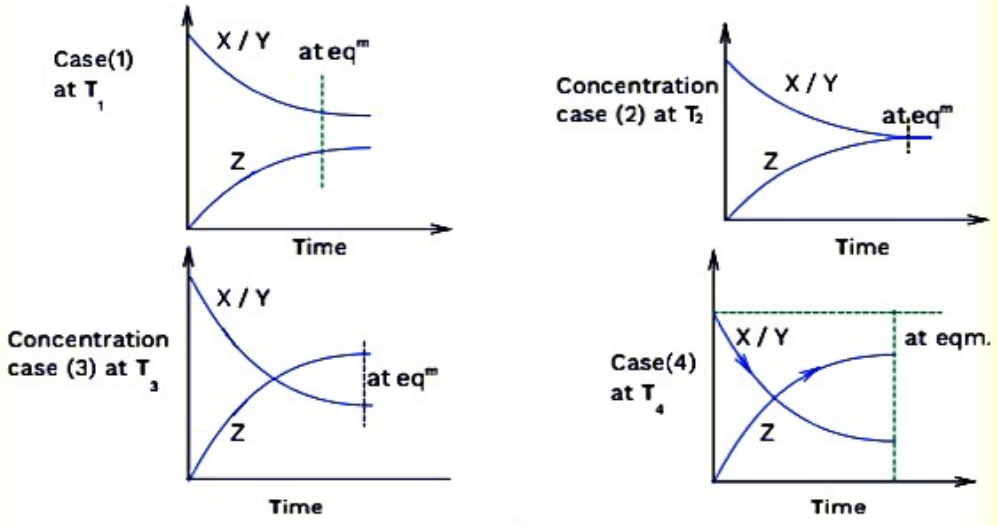
$$\text{Rate of Reaction} = -\frac{d[X]}{dt} = -\frac{d[Y]}{dt} = \frac{d[Z]}{dt} = k_f[X][Y] - k_b[Z]$$

At equilibrium R.O.R. = 0 $\Rightarrow -\frac{d[X]}{dt} = 0$, [X] = constant, [Y] = constant, [Z] = constant

$$K_c = \frac{k_f}{k_b} = \frac{[Z]_{eq.}}{[X]_{eq.}[Y]_{eq.}}$$

- **Graphs**

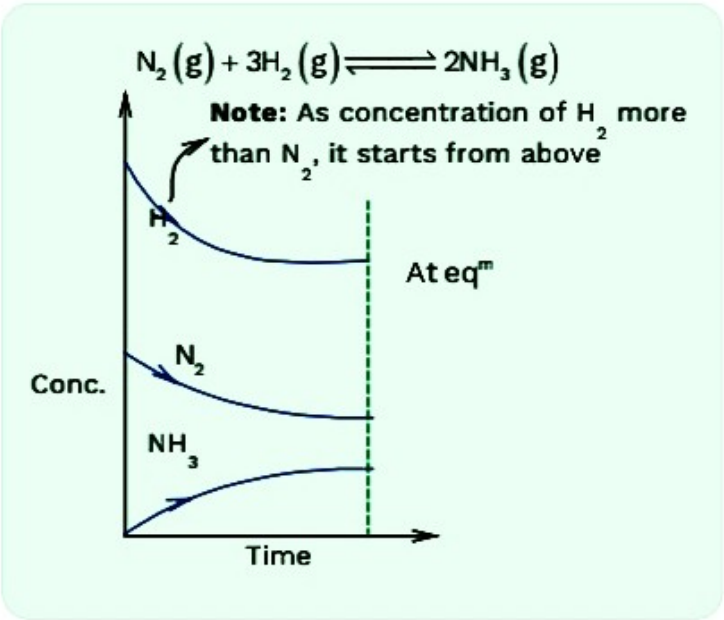
$$X(g) + Y(g) \rightleftharpoons Z(g)$$



% Conversion of reactant into prdouct

$4 > 3 > 2 > 1$

	$2HI(g) \rightleftharpoons$	$H_2(g)$	+	$I_2(g)$
Initial conc.	a	0		0
Conc. at eq.	a-2x	x		x

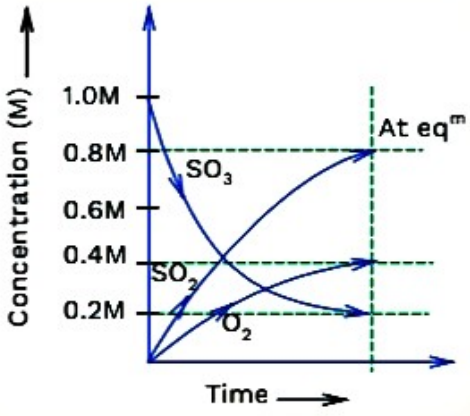


Example:

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$$

Initial	1M	0	0
Given at eqm.	1-2x	2x	x
	=0.2M	0.8M	0.4M

x = 0.4 M



Given: 80% conversion of reactant into product,

find K_c.

Sol. $K_C = \dfrac{[SO_2]^2[O_2]}{[SO_3]^2} = \dfrac{[0.8]^2[0.4]}{[0.2]^2} = 6.4$

Physical significance of the value of equilibrium constant

⇒ If the value of equilibrium constant is very high ($K_c > 10^3$), then reactants almost converted into products.

⇒ If the value of equilibrium constant is nearly 1 then appreciable conversion of reactants into products.

⇒ If the value of equilibrium constant is very low ($K_c < 10^{-3}$), then negligible conversion of reactants into products.

Equilibrium Constants

For gaseous reactions, $aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$

Equilibrium constant in terms of pressure (K_p)

$= \frac{P_C^c . P_D^d}{P_A^a . P_B^b}$; where P_i = Partial pressure of gas 'i' at equilibrium.

Equilibrium constant in terms of conentration(K_c)

$= \frac{[C]^c [D]^d}{[A]^a [B]^b}$; where $[A_i]$ = Concentration of A_i at equilibrium.

Equilibrium constant in terms of mole fraction (K_χ)

$= \frac{\chi_C^c \cdot \chi_D^d}{\chi_A^a \cdot \chi_B^b}$; where χ_i = Mole fraction of gas 'i' at equilibrium.

Units Of K_p And K_c

For $aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$

$$K_c = \frac{[C]_{eq.}^c [D]_{eq.}^d}{[A]_{eq.}^a [B]_{eq.}^b}$$ where [ ] ⇒ mol/litre.

$$\text{Unit Of } K_c = \frac{\left(\frac{mol}{lit}\right)^c \left(\frac{mol}{lit}\right)^d}{\left(\frac{mol}{lit}\right)^a \left(\frac{mol}{lit}\right)^b} = \left(\frac{mol}{lit}\right)^{(c+d)-(a+b)} = \left(\frac{mol}{lit}\right)^{\Delta n_g}$$

$$\text{Unit Of } K_p = \frac{(atm)^c (atm)^d}{(atm)^a (atm)^b} = (atm)^{(c+d)-(a+b)} = (atm)^{\Delta n_g}$$

Measurement of Partial Pressure

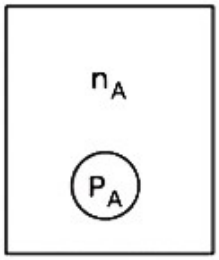
For ideal gas, $\boxed{PV = nRT}$

Equilibrium mixture:

$\boxed{\begin{array}{l} \text{Moles of A} = n_A \\ \text{Moles of B} = n_B \\ \text{Moles of C} = n_C \\ \text{Moles of D} = n_D \end{array}}$

$$\boxed{P_{total}V = n_{total}RT} \qquad ...(i)$$

where $n_{total} = n_A + n_B + n_C + n_D$



At constant T and V,

$$\boxed{P_AV = n_ART} \qquad ...(ii)$$

$$\frac{(ii)}{(i)} = \frac{P_A}{P_{total}} = \frac{n_A}{n_{total}} = \chi_A$$

$$P_A = \chi_A\left(P_{total}\right)$$

Partial pressure of gas A
= Mole fraction of gas A × Total pressure by gaseous mixture.

Relationship Between K_p and K_c

$$K_p = \frac{P_C^c \cdot P_D^d}{P_A^a \cdot P_B^b}, \qquad P_AV = n_ART,\ P_A = (n_A)RT/V,\ P_A = [A]RT$$

Therefore, $K_p = \dfrac{[C]^c[D]^d}{[A]^a[B]^b} \cdot (RT)^{(c+d)-(a+b)}$

$$\boxed{K_p = K_c(RT)^{\Delta n_g}}$$

Where, $\Delta n_g = (c+d) - (a+b)$

Δn_g = Sum of stoichiometric coefficient of all gaseous products - Sum of stoichiometric coefficient of all gaseous reactants.

$$P :\rightarrow \text{in atm},\ R = 0.821\frac{\text{atm L}}{\text{mol K}}$$

$[\ ]$ in mol / L

Examples: (How To Calculate Δn_g)

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ $\qquad \Delta n_g = (+1+1) - 1 = 1$

$CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$ $\qquad \Delta n_g = 1 - 1 - \frac{1}{2} = -\frac{1}{2}$

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ $\qquad \Delta n_g = 2$

Phase Equilibrium

$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$ $\quad \Delta n_g = 2$

$H_2O(g) \rightleftharpoons H_2O(l)$ $\quad \Delta n_g = -1$ (we consider gases only)

Relationship between K_p and K_χ

$$K_p = \frac{P_c^c P_d^d}{P_a^a P_b^b} = \frac{\chi_c^c \cdot \chi_d^d}{\chi_a^c \cdot \chi_b^b} P^{(c+c)-(a+b)}$$

As,

$$\boxed{P_A = \chi_A (P_{total})}$$

$$K_p = K_\chi P^{\Delta n_g}$$

$\Delta n_g = (c + d) - (a + b)$

If $\boxed{\Delta n_g = 0}$ then $\boxed{K_C = K_p = K_\chi}$

Types of Chemical Equilibrium

- **Homogeneous Chemical Equilibrium:**

All reactants and products are present in the same phase.

(1) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

$$K_p = \frac{P^2_{HI(g)}}{P_{H_2(g)} \cdot P_{I_2(g)}} \qquad K_c = \frac{[HI(g)]^2}{[H_2(g)][I_2(g)]}$$

(2) CH_3COOH (aq.)
CH_3COOH: Solute
$H_2O(l)$: Solvent

$$CH_3COOH(aq.) \rightleftharpoons CH_3COO^-(aq.) + H^+(aq.)$$

No K_p written for such reaction

$$K_c = \frac{[CH_3COO^-(aq.)][H^+(aq.)]}{[CH_3COOH(aq.)]}$$

(3) $NH_3(aq.) + \underset{\text{Solvent: excess}}{H_2O(l)} \rightleftharpoons NH_4^+(aq.) + OH^-(aq.)$

$$K_c = \frac{[NH_4^+(aq.)][OH^-(aq.)]}{[NH_3(aq.)]}$$

(Not $H_2O(l)$ is involved as it is solvent)

Concentration of solvent remains constant (does not change appreciably) during progress of reaction. Hence it is not considered in equilibrium constant expression.

(4) $CH_3COOC_2H_5(aq.) + H_2O(l) \rightleftharpoons CH_3COOH(aq.) + C_2H_5OH(aq.)$

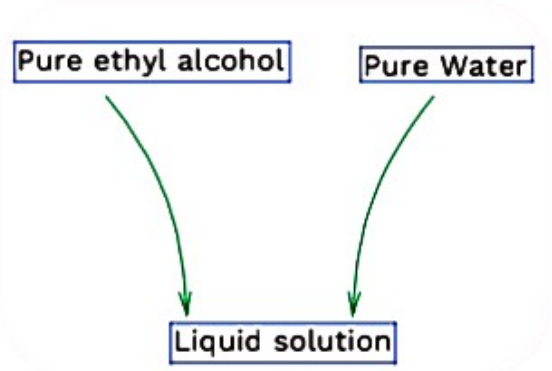
$$K_c = \frac{[CH_3COOH(aq.)][C_2H_5OH(aq.)]}{[CH_3COOC_2H_5(aq.)]}$$

(5) $CH_3COOC_2H_5(l) + H_2O(l) \rightleftharpoons CH_3COOH(l) + C_2H_5OH(l)$

$$K_c = \frac{[CH_3COOH(l)][C_2H_5OH(l)]}{[CH_3COOC_2H_5(l)][H_2O(l)]}$$

In this reaction H_2O (l) is not a solvent.

$$K_x = \frac{\chi_{CH_3COOH} \cdot \chi_{C_2H_5OH}}{\chi_{CH_3COOC_2H_5} \cdot \chi_{H_2O}}$$



Heterogeneous Chemical Equilibrium:

Reactants and products are present in more than one phase.

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

$K_p = P_{CO_2(g)}$ $\qquad$ $\boxed{K_c = [CO_2(g)]}$

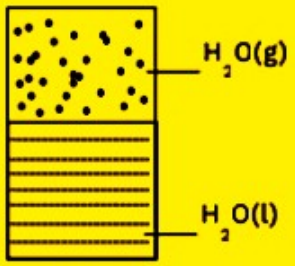
Density/Concentration of solid/pure liquid is constant during progress of reaction, hence they are not considered in equilibrium constant expression.

For solids: Density remains same
For pure liquids: Density remains same

$$\text{Molar Concentration (mol/L)} = \frac{\text{Density (g/L)}}{\text{Molar mass } (g/mol)}$$

↑ Therefore remains constant $\qquad$ ↓ (Always constant)

(1) Phase equilibrium



$H_2O(l) \rightleftharpoons H_2O(g)$

$K_p = P_{H_2O(g)}$

$K_c = [H_2O(g)]$

(2) $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

$K_p = P_{NH_3(g)} \cdot P_{H_2S(g)}$

(3) $NH_3(aq.) + \underset{\text{Solvent: excess}}{H_2O(l)} \rightleftharpoons NH_4^+(aq.) + OH^-(aq.)$

$$K_C = \frac{[NH_4^+(aq.)][OH^-(aq.)]}{[NH_3(aq.)]}$$

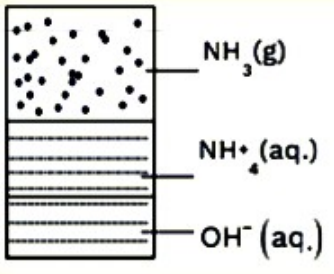
(4) $Ag_2O(s) + 2HNO_3(aq.) \rightleftharpoons 2AgNO_3(aq.) + H_2O(l)$

$$K_C = \frac{[AgNO_3(aq.)]^2}{[HNO_3(aq.)]^2}$$

When substances present in gaseous as well as in aqueous state, we don't write K_p or K_C, we write K_{PC}.

(1) $NH_3(g) + H_2O(l) \rightleftharpoons NH_4^+(aq.) + OH^-(aq.)$

$$K_{PC} = \frac{[NH_4^+(aq.)][OH^-(aq.)]}{P_{NH_3(g)}}$$



Concentration of $NH_3(g) = \frac{n}{V_{empty}}$

Concentration of $OH^-(aq) = \frac{n_{OH^-}}{V_{Sol}}$

(2) $2HCl(aq.) \rightleftharpoons H_2(g) + Cl_2(g)$

$$K_{PC} = \frac{P_{H_2(g)} P_{Cl_2(g)}}{[HCl(aq.)]^2}$$

(3) $Zn(s) + 2HCl(aq.) \rightleftharpoons ZnCl_2(aq.) + H_2(g)$

$$K_{PC} = \frac{[ZnCl_2(aq.)] P_{H_2(g)}}{[HCl(aq.)]^2}$$

Equilibrium Constant as per Thermodynamics:

Let: $aA(g) + bB(g) \rightleftharpoons cC(g)$

Reaction takes place at non-standard states.
$\Delta_r G = \Delta G^\circ + RT \ln Q$ → Reaction Quotient

At equilibrium,

$$\Delta_r G^\circ = -RT \ln K_{eq} = -2.303 RT \log_{10} K_{eq}$$
(no unit)

$$K_{eq} = \frac{a_C^c}{a_A^a a_B^b}$$

a=activity (no unit)

- For solid / pure liquid / solvent
  Activity = 1

  $a_i = 1$

- For ideal gases:

  $a_i = \frac{P_i}{P^\circ}$ where P_i = partial pressure of gas, P° = standard pressure

- For aqueous species:
  e.g.: CH_3COOH (aq.), OH^-(aq.)

  $a_i = \frac{Y_i \cdot [i]}{C^\circ}$; Y_i = activity coefficients (no unit)

- For dilute aqueous solution: $Y_i = 1$

  $a_i = \frac{[i]}{C^\circ}$

1. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$K_p^\circ = \frac{\left(\frac{P_{NH_3}}{P^\circ}\right)^2}{\left(\frac{P_{N_2(g)}}{P^\circ}\right)\left(\frac{P_{H_2(g)}}{P^\circ}\right)^3}$$

2. $HA(aq.) \rightleftharpoons H^+(aq.) + A^-(aq.)$

$$K_C^\circ = \frac{\left[\frac{[H^+(aq.)]}{C^\circ}\right]\left[\frac{[A^-(aq.)]}{C^\circ}\right]}{\left[\frac{[HA(aq.)]}{C^\circ}\right]}$$

3. $NH_3(g) + H_2O(l) \rightleftharpoons NH_4^+(aq.) + OH^-(aq.)$

$$K_{PC}^\circ = \frac{\left(\frac{[NH_4^+(aq.)]}{C^\circ}\right)\left(\frac{[OH^-(aq.)]}{C^\circ}\right)}{\left[P_{NH_3(g)} / P^\circ\right]}$$

Reacton Quotient (Q):

The reaction quotient(Q) measures the relative amounts of products and reactants present during a reaction at a particular point in time. The reaction quotient aids in figuring out which direction a reaction is likely to proceed, given either the pressures or the concentrations of the reactants and the products.

$aA(g) + bB(g) \rightleftharpoons cC(g)$

$$Q_p = \frac{P_C^c}{P_A^a P_B^b}, \qquad Q_C = \frac{[C(g)]^c}{[A(g)]^a [B(g)]^b}$$

P_A/[A(g)]: Pressure/concentration at some particular time not necessarily at equilibrium.

(1) If $Q < K_{eq}$ then reaction takes place in forward direction.
(2) If $Q = K_{eq}$ then reaction is at equilibrium.
(3) If $Q > K_{eq}$ then reaction takes place in backward direction.
This can be done by: ↓
(I) By adding C.
(Ii) By removing A and B.

Some facts about equilibrium constant:

Fact (1): **If we reverse a given chemical reaction then equilibrium constant of the new (reversed one) chemical reaction is reciprocal of the previous one.**

e.g $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ $\quad ...K_{eq.} = K_1$

$\therefore$ On reversing the above reaction we get:

$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ $\quad ...K_{eq.} = 1/K_1$

Explanation:

$$K_1 = \frac{[HI(g)]^2}{[H_2(g)][I_2(g)]}$$

$$\frac{1}{K_1} = \frac{[H_2(g)][I_2(g)]}{[HI(g)]^2}$$ which is equilibrium constant expression for reversed one chemical reaction.

Fact (2): **If we multiply a number to a existing chemical reaction then equilibrium constant of the new chemical reaction will be that number to be in power (indices) of the equilibrium constant of previous one.**

e.g $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $\quad ...K_{eq.} = K$

Multiply above reaction by 2

$2N_2(g) + 6H_2(g) \rightleftharpoons 4NH_3(g)$ $\quad ...K_{eq.} = K^2$

Explanation:

$$K_{eq.} = K = \frac{[NH_3(g)]^2}{[N_2(g)][H_2(g)]^3}$$

and $$K^2 = \frac{[NH_3(g)]^4}{[N_2(g)]^2[H_2(g)]^6}$$ which is equilibrium constant expression for new (multiplied one) chemical reaction.

Fact (3): **If we add two different chemical reactions, then equilibrium constant of the resultant chemical reaction will be product of equilibrium constants of added ones.**

e.g $A(g) + B(g) \rightleftharpoons C(g)$ $\quad ...K_{eq.} = K_1$

$D(g) \rightleftharpoons E(g)$ $\quad ...K_{eq.} = K_2$

Add above reactions:

$A(g) + B(g) + D(g) \rightleftharpoons C(g) + E(g)$ $\quad ...K_{eq.} = K_1.K_2$

Explanation:

$$K_1 = \frac{[C(g)]}{[A(g)][B(g)]}$$

$$K_2 = \frac{[E(g)]}{[D(g)]}$$

$$\therefore \quad K_1.K_2 = \frac{[C(g)]}{[A(g)][B(g)]} \cdot \frac{[E(g)]}{[D(g)]}$$

which is equilibrium constant expression for resultant added chemical reaction.

(iv) Unit of equilibrium constants;

$K_c, K_p, K_\chi \rightarrow$ may have unit

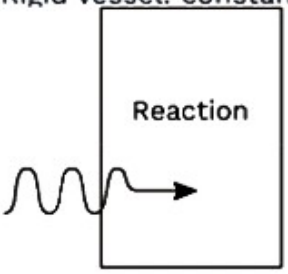
$K_c^\circ, K_p^\circ, K_\chi^\circ \rightarrow$ no units

Factors affecting equilibrium constant:

Temperature (governed by Vantt Hoff's equation)

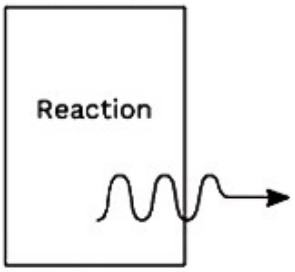
Some useful terminology:

(A) Rigid vessel: constant volume



Heat absorbed by reaction at constant volume is 80 J

$\boxed{\Delta U = 80J}$



Heat released at constant volume is 100 J

$\boxed{\Delta U = -100J}$

$Q_v = \Delta U$ Or ΔE

U: Internal energy

ΔU: Change in internal energy.

(B) At Constant pressure:
$Q_p = \Delta H$
H : Enthalpy
ΔH: Change in enthalpy
Heat released at constant pressure is 50 J
ΔH = – 50 J
And heat absorbed at constant pressure is 100 J.
ΔH = +100 J

Types Of Reactions:

(I) Endothermic reactions:
Heat absorbed in reactions:

$$2A(g) \rightleftharpoons B(g) + C(g),\ \Delta H = y\,kJ, \Delta_r H^\circ = y\,\frac{kJ}{mole}$$

$$A(g) \rightleftharpoons \frac{B}{2}(g) + \frac{C}{2}(g),\ \Delta H = \frac{y}{2}\,kJ$$

(II) Exothermic reactions:
Heat released in reactions:

$$2A(g) \rightleftharpoons B(g) + C(g),\ \Delta H = -y\,kJ, \Delta_r H^\circ = -y\,\frac{kJ}{mole}$$

$$A(g) \rightleftharpoons \frac{B}{2}(g) + \frac{C}{2}(g),\ \Delta H = -\frac{y}{2}\,kJ$$

Vant Hoff's Equation:

$$\ln\left(\frac{K_2}{K_1}\right) = \frac{\Delta_r H^\circ}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

R = 8.314 J/mol K
$\Delta_r H^\circ$ = Enthalpy of reactions at standard states.
K_1= Equilibrium constant at temperature T_1.
K_2= Equilibrium constant at temperature T_2.

For endothermic reactions

$\Delta_r H^\circ$ = +ve
Equilibrium constant increases with increase in temperature.
Temperature (↑), K_{eq}(↑) as clear from Vantt Hoff's equation.
Means more conversion of reactants into products at higher temperature for such reactions.

In exothermic reactions:

$\Delta_r H^\circ$ = –ve
Equilibrium constant decreases with increase in temperature.
Temperature (↑), K_{eq}(↓) as clear from Vantt Hoff's equation.
Means less conversion of reactants into products at higher temperature for such reactions.

Calculation of K_p and K_c

Stoichiometric method: (Applicable for all type of reaction)

$$aA(g) \rightleftharpoons bB(g) + cC(g)$$

a	0	0	:	initial moles
a – ax	bx	cx	:	final moles at equilibrium

(i) $$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$$

Initial:	a	0	0
At eqm.:	a – 2x	x	x
Conc. at eqm.:	$\frac{a-2x}{V}$	$\frac{x}{V}$	$\frac{x}{V}$

(V : Volume of vessel)

Total moles at equilibrium = a
Partial pressure of H_2 = mole fraction (H_2) × Total pressure

$$P_{H_2} = \frac{x}{a} \cdot P_{total}$$

$$K_c = \frac{[H_2(g)][I_2(g)]}{[HI(g)]^2} = \frac{\left(\frac{x}{V}\right)^2}{\left(\frac{a-2x}{V}\right)^2} = \frac{x^2}{(a-2x)^2}$$

$$K_p = \frac{P_{H_2} P_{I_2}}{(P_{HI})^2} = \frac{\left(\frac{xP_{total}}{a}\right)^2}{\left(\frac{(a-2x)P_{total}}{a}\right)^2} = \frac{x^2}{(a-2x)^2}$$

$\therefore$ $K_p = K_c$ as $\Delta n_g = 0$

(ii) $$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Initial:	a	b	0
Moles at eqm.:	a-x	b – 3x	2x
Conc. at eqm.:	$\frac{a-x}{V}$	$\frac{b-3x}{V}$	$\frac{2x}{V}$

Total moles at quilibrium = a + b − 2 x

$$K_C = \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{a-x}{V}\right)\left(\frac{b-2x}{V}\right)^3} = \frac{4x^2V^2}{(a-x)(b-2x)^3}$$

$$K_P = \frac{\left(P_{NH_3}\right)^2}{P_{N_2}\left(P_{H_2}\right)^3} = \frac{\left(\left(\frac{2x}{a+b-2x}\right)P\right)^2}{\left(\frac{a-x}{a+b-2x}P\right)\left(\frac{b-3x}{a+b-2x}P\right)^3} = \frac{4x^2(a+b-2x)^2}{(a-x)(b-3x)^3P^2}$$

Also for above rection:

$K_p = K_C(RT)^{\Delta n_g}$

$\Delta n_g = -2$ (above)

For ideal gases:

Volume % = Mole % at constant Pressure and Temperature.

At equilibrium $NH_3(g)$ is 25% by volume (let given)

$$\therefore \quad \frac{2x}{a+b-2x}.100 = 25$$

Note: If 'c' moles of inert gas are also present in vessel at equilibrium

Total moles at equilibrium = a + b + c − 2x

$$P_{total} = P_{H_2} + P_{NH_3} + P_{N_2} + P_{inert\,gas}$$

Now, partial pressure of $H_2 = \frac{(b-3x)}{(a+b+c-2x)}P_{total}$

Degree of Dissociation (DOD)

It is fraction of amount (generally moles) of reactants converted into products.

Ideally this method is applicable for dissociation reactions.

DOD is generally represented by 'α'.

and $\boxed{\%\ DOD = \alpha \times 100}$

Suppose $A(g) \rightleftharpoons 2B(g)$

Initial moles:	a	0
At eqm. Moles:	a – x	2x

$$\therefore \quad \alpha = \frac{\text{Moles of reactant converted}}{\text{Initial moles of reactant}}$$

$$\boxed{\alpha = \frac{x}{a}} \quad \text{or} \quad \boxed{\%\alpha = \frac{100x}{a}}$$

Application

$$A(g) \rightleftharpoons B(g) + 2C(g)$$

Initial moles:	M	0	0
Moles at eq.:	M–Mα	Mα	2Mα

To check → $\boxed{\alpha = \frac{M\alpha\,(\text{Reacted amount})}{M\,(\text{Initial amount})} = \alpha}$

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$$

Initial moles:	a	0	0
moles at eq.:	a – x	x/2	x/2
Or moles at eq.:	a (1– α)	aα/2	aα/2

where $\boxed{\alpha = \frac{x}{a}}$

$$K_c = \frac{[H_2][I_2]}{[HI]^2}$$

$$K_C = \frac{\left(\frac{a\alpha}{2V}\right)^2}{\left(\frac{a(1-\alpha)}{V}\right)^2}$$ where V is volume of container

$$\boxed{K_C = \frac{\alpha^2}{4(1-\alpha)^2}}$$

Degree of dissociation in terms of molar mass / vapour density:

Suppose dissociation reaction:

$$A(g) \rightleftharpoons \underbrace{B(g) + C(g) + D(g) + ...}_{\text{'n' moles}}$$

or; $A(g) \rightleftharpoons nB(g)$

Initial moles:	a	0
Moles at eq.:	$a(1-\alpha)$	$na\alpha$

$$\frac{\text{Total moles at equilibrium}}{\text{Initial moles of reactant}} = \frac{a(1-\alpha)+na\alpha}{a} = \frac{1+(n-1)\alpha}{1} = \frac{\frac{W}{M_o}}{\frac{W}{M_T}}$$

(As mass is conserved)

$$\frac{1+(n-1)\alpha}{1} = \frac{M_T}{M_o}, \text{ where}$$

M_T : Theoretical molar mass of reactant.
M_o: Observed molar mass or average molar mass of mixture.
n : No. Of moles of gaseous products per mole of reactant.

$\therefore$ Degree of dissociation, $\boxed{\alpha = \frac{M_T - M_o}{M_o(n-1)}}$ For $\boxed{\Delta n_g > 0}$

e.g. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

M_T = molar mass of PCl_5.
M_o = molar mass of mixture containing PCl_5, PCl_3, Cl_2.

DOD in terms of vapour density:

$$\boxed{\alpha = \frac{D_T - D_o}{D_o(n-1)}}$$

D_T = Vapour density of reactant = $M_T/2$
D_o = Vapour density of mixture = $M_o/2$
Just put them in $\alpha = \frac{M_T - M_O}{M_O(n-1)}$

We will get the above expression.

Calculation of (n): No. Of moles of gaseous products per mole of reactant.

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$; n = 2

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$; n = 4/2 = 2

Note: • $\boxed{0 < \alpha < 1}$ always for reversible reaction.

• $\frac{M_T}{M_o} > 1$, for dissociation reactions.

$\therefore \quad M_T > M_o$

As, Moles = W/M

$$M_{mix} = \frac{W(\text{constant})}{n_{mix.}}$$

Degree of association:

$$nA(g) \rightleftharpoons A_n(g)$$

Initial moles:	a	0
Moles at eq.:	a (1 - α)	aα/n

$$\frac{\text{Total moles at equilibrium}}{\text{Initial moles of reactant}} = \frac{a(1-\alpha) + \frac{a\alpha}{n}}{a} = \frac{1 + \left(\frac{1}{n} - 1\right)\alpha}{1} = \frac{\frac{W}{M_c}}{\frac{W}{M_T}} = \frac{M_T}{M_c}$$

Degree of association, $\boxed{\alpha = \frac{M_T - M_c}{M_o\left(\frac{1}{n} - 1\right)}}$

Note:

For dimerization, n = 2 and Trimerization, n = 3 and so on....

For association, $\Delta n_g < 0$

$0 < \alpha < 1$

$\frac{M_T}{M_o} < 1$

$M_O > M_T$

e.g. Trimerization of Se_2:

$$3Se_2(g) \rightleftharpoons Se_6(g)$$

Illustrations

Q 1.: **For the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$. The equilibrium mixture was found to contain 2 moles of N_2O_4 and 1 mole of NO_2 in 10 L vessel at 300 K. Calculate K_p (in atm)**

Sol.: $$K_c = \frac{[NO_2(g)]^2}{[N_2O_4(g)]} = \frac{(1/10)^2}{(2/10)} = \frac{1}{20}\,M$$

$$K_p = K_C(RT)^{\Delta n_g}, \qquad \Delta n_g = 1(\text{Here})$$

$$\therefore \quad K_P = \frac{1}{20}(0.0821 \times 300)^1 = 1.23 \text{ atm}$$

Q 2.: **At constant temperature, equilibrium constant K_c is 16 for the reaction:**
$A(g) + B(g) \rightleftharpoons C(g) + D(g)$
If we take 2 moles of each four gas in a 10 L Vessel. What would be equilibrium concentration of B and C?

Sol.: $$Q_c = \frac{(2/10)^2}{(2/10)^2} = 1$$

But $K_c = 16$
And $K_c > Q_c$ means reaction proceeds in forward direction.

$$A(g) + B(g) \rightleftharpoons C(g) + D(g)$$

At equilibrium: 2-x 2-x 2+x 2+x

$$\frac{(2+x)^2}{(2-x)^2} = 16$$

$$\frac{2+x}{2-x} = 4$$

$2 + x = 4(2 - x)$
$5x = 6$
$x = 1.2$

Now concentration of B $= \frac{2-1.2}{10} = 0.08\,M$

Concentration of C $= \frac{2+1.2}{10} = 0.32\,M$

Q 3.: **A mixture of SO_3, SO_2, O_2 gases is maintained in a 10 L vessel.**
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$; $K_c = 100$
If the number of moles of SO_2 and SO_3 in flask are equal at equilibrium, then calculate number of moles of O_2.

Sol.: $$K_C = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$$

$$100 = \frac{1}{[O_2]} \quad \text{...[Given: } n_{SO_2} = n_{SO_3} \text{ at equilibrium]}$$

$$[O_2] = \frac{1}{100} = \frac{n_{O_2}}{V}$$

$$\therefore \quad \boxed{n_{O_2} = 0.1}$$

Q 4.: **If $2A(g) \rightleftharpoons B(g) + C(g)$ reaction starts with only A. At equilibrium volume % of gases A and B is 60. Calculate K_p if total pressure at equilibrium is 10 atm.**

Sol.: For ideal gases
Volume % = mole % at constant P & T.

$$2A(g) \rightleftharpoons B(g) + C(g)$$

Initial moles:	a	0	0
At equilibrium:	a – 2x	x	x

Given:

$$\frac{\text{Mole of A and B at equilibrium}}{\text{Total moles at equilibrium}} \times 100 = \frac{(a - 2x + x)}{a} \times 100 = 60$$

$$\Rightarrow \quad (a - x)100 = 60a$$

$$\Rightarrow \quad 100a - 100x = 60a$$

$$\Rightarrow \quad \frac{40}{100} = \frac{x}{a} = 0.4$$

$$K_p = \frac{\left(\frac{x}{a}\right)^2 \times (10)^2}{\left(\frac{a - 2x}{a}\right)^2 \times 10^2}$$

$$\Rightarrow \quad \frac{(0.4)^2}{(1 - 2 \times 0.4)^2} = 4$$

Q 5.: Calculate mole% of chlorine at equilibrium under a total pressure of 2 atm. Given that:

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; $K_p = 0.2$

Sol.: Given: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; $K_p = 0.2$

Initial moles:	a	0	0
At equilibrium:	a – x	x	x

$$K_p = \frac{\left(\frac{x}{a+x}\right)^2 \times 4}{\left(\frac{a-x}{a+x}\right) \times 2}$$

$$\Rightarrow \quad 0.2 = \frac{\left(\frac{x}{a+x}\right)^2 \times 4}{\left(\frac{a-x}{a+x}\right) \times 2}$$

$$\Rightarrow \quad 0.2 = \frac{2x^2/(a+x)}{a-x}$$

$$\Rightarrow \quad a^2 - x^2 = 10x^2$$

$$x = \frac{a}{\sqrt{11}}$$

$$\text{Mole\% Of } Cl_2 = \frac{x}{a+x} \times 100$$

$$= \frac{\frac{a}{\sqrt{11}}}{a\frac{\left(\sqrt{11}+1\right)}{\sqrt{11}}} = \frac{1}{\sqrt{11}+1} \times 100$$

Q 6.: The degree of dissociation (α) is 0.4 at 400 K and 1 atm for gaseous reaction: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. Find density of mixture of gases at equilibrium.

Sol.: Given: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Initial moles:	a	0	0	
At equilibrium:	$a(1-0.4)$	0.4a	0.4a	as $\alpha = 0.4$, $n = 2$

Apply

$$0.4 = \frac{D_T - D_O}{D_O (n-1)}$$

M_T = molar mass of PCl_5 = 31 + 35.5 × 5 = 208.5

Or, $$0.4 = \frac{M_T - M_O}{M_O (n-1)}$$

$$0.4 = \frac{208.5 - M_o}{M_o (1)}$$

M_O = 149 gm

Required Density $= \frac{P \times M_O}{RT} = 4.53\,gm/L$

Q 7.: Calculate equilibrium concentration of PCl_3 in a 10 L rigid vessel if 1 mol of PCl_5 is taken to attain equilibrium according to gaseous reactions.

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(i) If $K_c = 10^{-5}$ at temperature T_1.

(ii) If $K_c = 10^5$ at temperature T_2.

(iii) If $K_c = 10^{-2}$ at temperature T_3.

Sol.: (i) Very less conversion of reactants into products.

Given: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Equi. moles: $1-x \approx 1$ x x

Solving $$K_C = \frac{(x/10)^2}{(1/10)} \Rightarrow 10^{-5} = \frac{x^2}{10}$$

$\Rightarrow x^2 = 10^{-4}$

$\Rightarrow x = 10^{-2}$

$\therefore [PCl_3] = \frac{x}{10} = 10^{-3} M$

[If x < 5% of initial value then our assumption is valid]

(ii) Almost all reactants converted into products.

Given: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Equi. moles: $1-x$ x x

At equilibrium: y $x \approx 1$ $x \approx 1$

$$10^5 = \frac{\left(\frac{1}{10}\right)^2}{\left(\frac{y}{10}\right)} \Rightarrow y = 10^{-6}$$

$1 - x \approx y$

$1 - x \approx 10^{-6}$

$x \simeq 1$

$$\left[PCl_3\right] = \frac{x}{10} \simeq \frac{1}{10}$$

$$\left[PCl_5\right] = \frac{y}{10} = 10^{-7}$$

(iii) Given: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Equi. moles: 1-x x x

$$K_c = \frac{\frac{(x/10)^2}{(1-x)}}{10}$$

$$10^{-2} = \frac{x^2}{10(1-x)}$$

$$x^2 = 0.1 - 0.1x$$

$$10x^2 + x - 1 = 0$$

$$x = \frac{\sqrt{41} - 1}{20}$$

$$\left[PCl_3\right] = \frac{x}{10} = \frac{\sqrt{41} - 1}{200} = 0.027M$$

Le Chatelier's Principle

When reaction conditions are changed, then equilibrium shifts in that direction where effect of change is nullify on minimise.

Suppose a gaseous chemical reaction:

$aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$

We can write:

$$K_C = \frac{[C]^c[D]^d}{[A]^a[B]^b} = \frac{n_C^c . n_D^d}{n_B^b . n_A^a}\left(\frac{1}{V}\right)^{\Delta n_g}$$

$$K_p = \frac{P_C^c P_D^d}{P_A^a P_B^b} = \frac{n_C^c . n_D^d}{n_A^a . n_B^b}\left(\frac{P}{n_{total}}\right)^{\Delta n_g}$$

$$\because P_C = \frac{n_c}{n_{total}} \times P$$

Case I: No. of moles/concentration (If volume is constant)

$\uparrow [A] = n_A \uparrow$

V – (Constant)

(A) On addition of reactants/removal of products:-
Equilibrium shifts in forward direction.

(B) On removal of reactants/addition of products:-
Equilibrium shifts in backward direction.

NOTE: Above actions are not valid, if reactant or product is **solid**.

Case II: Effect of Pressure:

(At constant temperature and moles)

As for ideal gases: $P \propto \frac{1}{V}$

Case A: On increasing pressure (volume decreases)

(1) If $\Delta n_g = 0$, no effect.

(2) If $\Delta n_g > 0$, equilibrium shifts in **Backward** direction.

(3) If $\Delta n_g < 0$, equilibrium shifts in **Forward** direction.

*** Conclusion:**
When we increase pressure, equilibrium shifts towards less number of gaseous moles in reaction.

Case B: On decreasing pressure(volume increases)

(1) If $\Delta n_g = 0$, no effect.

(2) If $\Delta n_g > 0$, equilibrium shifts in **Forward** direction.

(3) If $\Delta n_g < 0$, equilibrium shifts in **Backward** direction.

*** Conclusion:**
When we decreases pressure, volume increases, equilibrium shifts towards more number of gaseous moles in reaction.

e.g $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(On decreasing pressure, equilibrium goes forward)

Case III: CONCENTRATION (effect by pressure)

* When we increase pressure (decrease volume) concentration of reactants and products increases.

New equilibrium concentration > Old equilibrium concentration.

	A(g)	+	B(g)	$\rightleftharpoons$	C(g)	
At equi. (V=volume of vessel)	1		1		1	$\therefore K_c = 1$
At new conditon $\left(\frac{V}{2}=\text{volume of vessel}\right)$	2		2		2	$\therefore Q_c = \frac{1}{2}$

Means reaction proceeds in forward direction

At new equilibrium	$2-x$	$2-x$	$2+x$
	>1	>1	>1

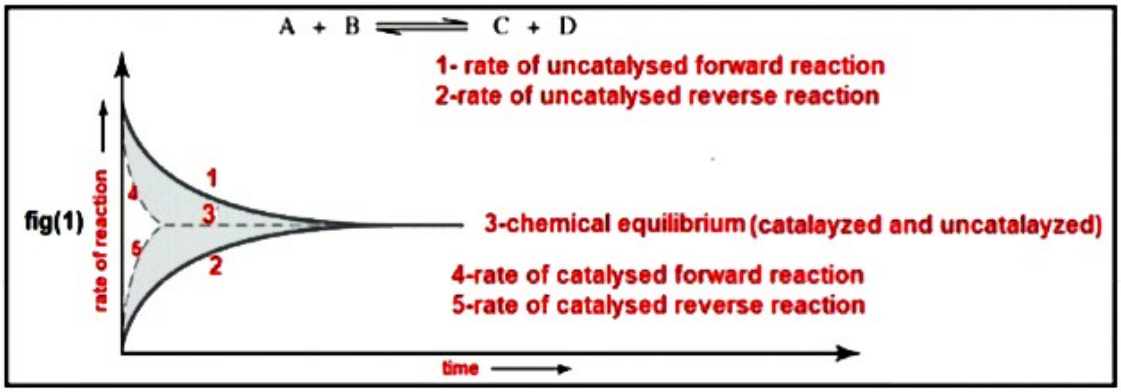
$(\because x \in (0,1))$

As at $x = 1$
$K_c = 3$ which is not possible

Case IV: CATALYST ADDITION:

No effect.

* Catalyst increases forward and backward reaction upto same extent, so equilibrium concentration remains same.



Case V: **Addition of INERT GAS**: (Non-Reactive)

(A). Addition of inert gas at constant volume: [rigid vessels]

→ Molar concentration of every substance remain same.

As inert gas does not react with any reactants or products.

So, **No effect.**

A, B, C, D, +E (Inert gas) at constant T and V, $P \propto n$

Partial Pressure of A $= \frac{n_A}{n_{total}} \times P \uparrow = \text{constant}$

(B). Addition of inert gas at constant pressure:

(1) If $\Delta n_g = 0$, no effect.

(2) If $\Delta n_g > 0$, equilibrium shifts in **Forward** direction.

(3) If $\Delta n_g < 0$, equilibrium shifts in **Backward** direction.

Explanation Thinking:

(using formula of K_C and K_P)

(and keep values of K_C, K_P constant)

Conclusion:-

When we add inert gas, at constant pressure equilibrium shifts towards more number of gaseous moles in reaction.

Case VI: Effect of Temperature

$$\ln\left(\frac{K_2}{K_1}\right) = \frac{\Delta_r H}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$ **:- Vantt Hoff's Equation**

(A) For Endothermic Reactions:

$\Delta_r H$ = +ve

(i) On increasing temperature, equilibrium constant increases, so equilibrium shifts in **forward** direction.

(ii) On decreasing temperature, equilibrium constant decreases, so equilibrium shifts in **backward** direction.

Remember:
Increase in temperature, favours endothermic reactions.

(B) For Exothermic Reactions:

$\Delta_r H$ = –ve

(i) On increasing temperature, equilibrium constant decreases for such reactions, so equilibrium shifts in backward direction. (less conversion of reactants into products)

(ii) On decreasing temperature, equilibrium constant increases for such reactions, so equilibrium shifts in forward direction. (more conversion of reactants into products)

Above statements can be easily verified using Vantt Hoff's equation.

Q1.: **What are the favourable conditions for the dissociation of ammonia in the reaction given below?**

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ $\quad \Delta_r H$ = –ve

Sol.: Favourable condition for dissociation of NH_3

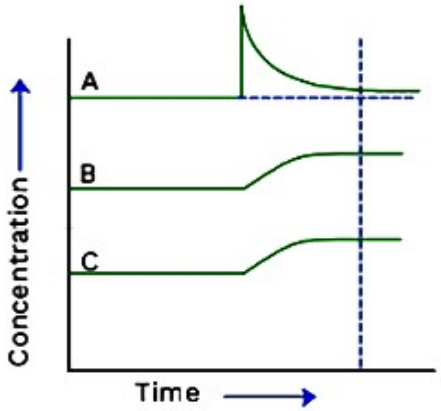
(1) Removal of N_2 or H_2
(2) Low pressure.
(3) High temperature as given reaction is exothermic.
(4) Addition of inert gas at constant pressure.

Important Note:

	$A(g) + B(g) \rightleftharpoons 2C(g)$		
Concentration at equilibrium	a	b	c
On addition of C			
Concentration at given condition	a	b	c + d
Concentration at new equilibrium	(a + x)	(b + x)	(c + d) – 2x
	>a	>b	>c
			< c + d

$Q_c > K_c$ equilibrium shifts in backward direction.

New equilibrium condition:



Simultaneous Equilibria

When two or more than two equilibrium simultaneously develop in a closed vessel then the situation is known as simultaneous equilibrium.

* How to write concentration in simultaneous equilibrium:

$$A(g) \rightleftharpoons 2B(g) + C(g)$$

1. Let initial moles of A = 1

At equilibrium 1-x (2x-y) x

$$B(g) \rightleftharpoons 3D(g)$$

At equilibrium (2x-y) 3y

A, B, C, D Moles of B is constant in both because of homogenous mixture.

$n_{total} = n_A + n_B + n_C + n_D = (1-x) + (2x-y) + x + 3y$

$= 1 + 2x + 2y$

2. $$2A(g) \rightleftharpoons B(g) + 3C(g)$$

At equilibrium: $(1-2x)$ x $(3x-y)$

$$C(g) \rightleftharpoons 2E(g)$$

At equilibrium: $(3x-y)$ 2y

3. $$A(g) \rightleftharpoons 2B(g) + C(g)$$

At equilibrium: (1−x) (2x-y) (x + 3y)

$$B(g) \rightleftharpoons 3C(g) + D(g)$$

At equilibrium: (2x−y) (3y+x) y

4. $$A(g) + 2B(g) \rightleftharpoons 3C(g)$$

At equilibrium: (a−x−y) (b−2x) 3x

$$A(g) \rightleftharpoons 2D(g)$$

At equilibrium: a − x − y 2y

Q1.: When NO and NO_2 are mixed, the following equilibriums are obtained:

$2NO_2(g) \rightleftharpoons N_2O_4(g)$ $K_p = 6.8$ atm

$NO(g) + NO_2(g) \rightleftharpoons N_2O_3(g)$ $K_P = ?$

In an experiment when NO and NO_2 are mixed in ratio 1 : 2, the $P_{total} = 5.05$ atm and $P_{N_2O_4} = 1.7$ atm at equilibrium.

Calculate:-

(i) The equilibrium Partial pressure of NO.

(ii) K_p for $NO(g) + NO_2(g) \rightleftharpoons N_2O_3(g)$

Sol.: $2NO_2(g) \rightleftharpoons N_2O_4(g) \quad K_p = 6.8 \text{ atm}$

At equilibrium $(2P-2y-x) \quad y$

Let initial $P_{NO} = P$, then $P_{NO_2} = 2P$

$$NO(g) + NO_2(g) \rightleftharpoons N_2O_3(g) \quad K_p = ?$$

At equilibrium $(P-x) \quad (2P-2y-x) \quad x$

1. $$6.8 = \frac{P_{N_2O_4}}{(P_{NO_2})^2} = \frac{1.7}{(P_{NO_2})^2}$$

$$\boxed{P_{NO_2} = 0.5}$$

$\therefore \quad 2P - 2y - x = 0.5 \qquad (y = 1.7)$

$$\boxed{2P - x = 3.9}$$

$$P_{total} = P_{NO_2} + P_{N_2O_4} + P_{NO} + P_{N_2O_3} = 0.5 + 1.7 + (P - x) + x$$

$5.05 = 2.2 + P$

$P = 2.85$

$2P - x = 3.8$

$2(2.85) - x = 3.8$

$\boxed{x = 1.8}$

(i) $P_{NO} = P - x = 2.85 - 1.8 = \boxed{1.05 \text{ atm}}$

(ii) $$K_p = \frac{P_{N_2O_3}}{P_{NO} \cdot P_{NO_2}} = \frac{1.8}{1.05 \times .5} = 3.42 \text{ atm}$$

Q2.: Given equilibrium total pressure = P. Find K_p in terms of P

Sol.: $A(s) \rightleftharpoons B(g) + 2C(g)$

At equilibrium $\quad P/3 \quad 2P/3$

$$K_p = \left(\frac{P}{3}\right)\left(\frac{2P}{3}\right)^2 = \boxed{\frac{4P^3}{9}}$$

Q3.: **At constant temperature and volume:**

	$X(s) \rightleftharpoons$	$A(g) + B(g)$	
At equilibrium	–	$(x + y)$	x
	$Y(s) \rightleftharpoons$	$A(g) + C(g)$	
At equilibrium	–	$(x + y)$	y

In same vessel

$P_{total} = P$

Find relationship between x and y and different equilibrium constants.

Sol.: $P_{total} = P_A + P_B + P_C = 2(x + y)$

x = Partial pressure of gas B when both equation simultaneously developed.
y = Partial pressure of gas C when both equation simultaneously developed.

$K_{P_1} = (P_A)_{total} P_B = (x + y)x$ ...(1)

$K_{P_2} = (P_A)_{total} P_C = (x + y)y$ ...(2)

$$\frac{(1)}{(2)} = \boxed{\frac{P_B}{P_C} = \frac{x}{y} = \frac{K_{P_1}}{K_{P_2}}}$$

$(1) + (2) = (x + y)^2 = K_{P_1} + K_{P_2}$

$$\boxed{P_A = \sqrt{K_{P_1} + K_{P_2}} = x + y}$$

Case 2: **If equilibrium are established in different vessels.**

		$X(s) \rightleftharpoons A(g) + B(g)$	
At equilibrium	–	$\frac{P_1}{2}$	$\frac{P_1}{2}$

Equilibrium Pressure = P_1

$$\boxed{K_{P_1} = \left(\frac{P_1}{2}\right)^2}$$

		$Y(s) \rightleftharpoons A(g) + B(g)$	
At equilibrium	–	$\frac{P_2}{2}$	$\frac{P_2}{2}$

Equilibrium Pressure = P_2

$$\boxed{K_{P_2} = \left(\frac{P_2}{2}\right)^2}$$

Now if these different vessels are mixed, total pressure of $A = \frac{P_1}{2} + \frac{P_2}{2}$ then their individual Q_{P_1} and Q_{P_2} is more than K_{P_1} and K_{P_2}, therefore reaction goes in backward direction. Now reaction becomes as case (1)

(where x and y) are coefficients.

where $\boxed{x < \frac{P_1}{2} \text{ and } y < \frac{P_2}{2}}$.

Phase Equilibrium

A phase is a homogeneous portion of a system.

Phase equilibria are always heterogenous.

$$X(s) \underset{\text{freezing}}{\overset{\text{Melting / fusion}}{\rightleftharpoons}} X(l)$$

MELTING POINT (FREEZING POINT):

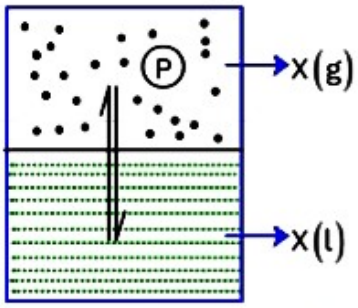
Temperature at which Rate of melting of solid X = Rate of freezing of X(l).

If external pressure = 1 atm, then this temperature is called normal melting point / freezing point.

If external pressure = 1 bar, then this temperature is called standard melting point / freezing point.

$$X(\ell) \rightleftharpoons X(g)$$

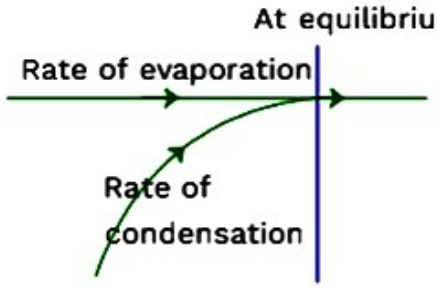
P = Vapour pressure of liquid $X(\ell)$



Rate of evaporation = constant (depends as surface area.)

Rate of condensation increases with time.

Rate of condensation depends upon number of particles per unit volume in vapour pressure.



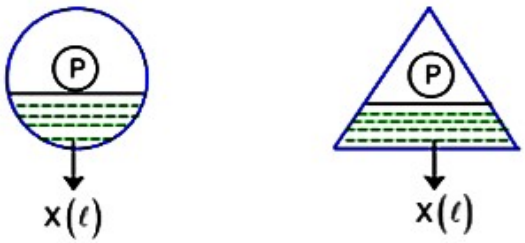
At equilibrium: Rate of evaporation of X(l) = Rate of condensation of X(g)

Vapour pressure:- Pressure exerted by vapour on its liquid surface at equilibrium is called vapour pressure of liquid.

Vapour pressure doesn't depend on size or shape of vessel, not a

surface phenomena.

At constant temperature and for same liquid.



Vapour pressure of pure liquid doesn't depend on amount of pure liquid present at equilibrium. Because at constant T,

$\boxed{K_p = P_{X(g)}}$ (No need of X(l))

$\boxed{K_c = [X(g)]}$

Vapour pressure depends on temperature.

$X(l) \rightleftharpoons X(g)$ $\quad \boxed{\underset{\text{vapourisation}}{\Delta H = +ve}}$

On increasing temperature, vapour pressure increases

At $T_1 : K_{P_1} = P_1$ (vapour pressure of X(l) at T_1)

At $T_2 : K_{P_2} = P_2$ (vapour pressure of X(l) at T_2)

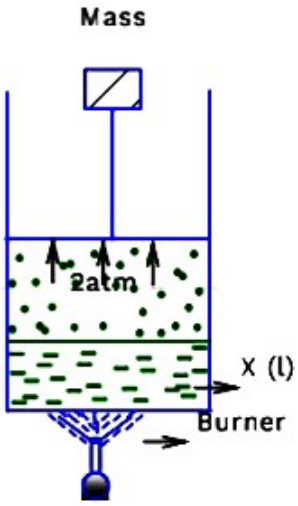
Using formula:

$$\boxed{\ln\left(\frac{P_2}{P_1}\right) = \frac{\Delta H_{vap.}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$$

More volatile substance has more vapour pressure and low boiling point.

$$\boxed{\%\,\text{Relative Humidity} = \frac{\text{Partial pressure of } H_2O(g)}{\text{Vapour pressure of } H_2O(l)} \times 100}$$

Boiling point: Temperature at which vapour pressure of liquid becomes equal to external pressure.



P_{ext} = 2 atm

Then Vapour pressure = 2 atm

On increasing Pressure, boiling point increases

If external pressure = 1 atm, then it is normal boiling point.
If external pressure = 1 bar, then it is standard boiling point.

$H_2O(s) \rightleftharpoons H_2O(l)$, ΔV = –ve, volume decreases, ΔH_{fusion} = +ve

$\Delta V = [V_m(H_2O(l)) - V_m(H_2O(s))]$

Universal Law:
When pressure increases, specific volume decreases.
Specific volume = Volume per unit mass.

Favourable Condition for Melting of Ice
High pressure, high temperature.

For General Substances:

$X(s) \rightleftharpoons X(l)$ ΔV = +ve volume increases

$V(X(l)) > V(X(s))$; ΔH = +ve

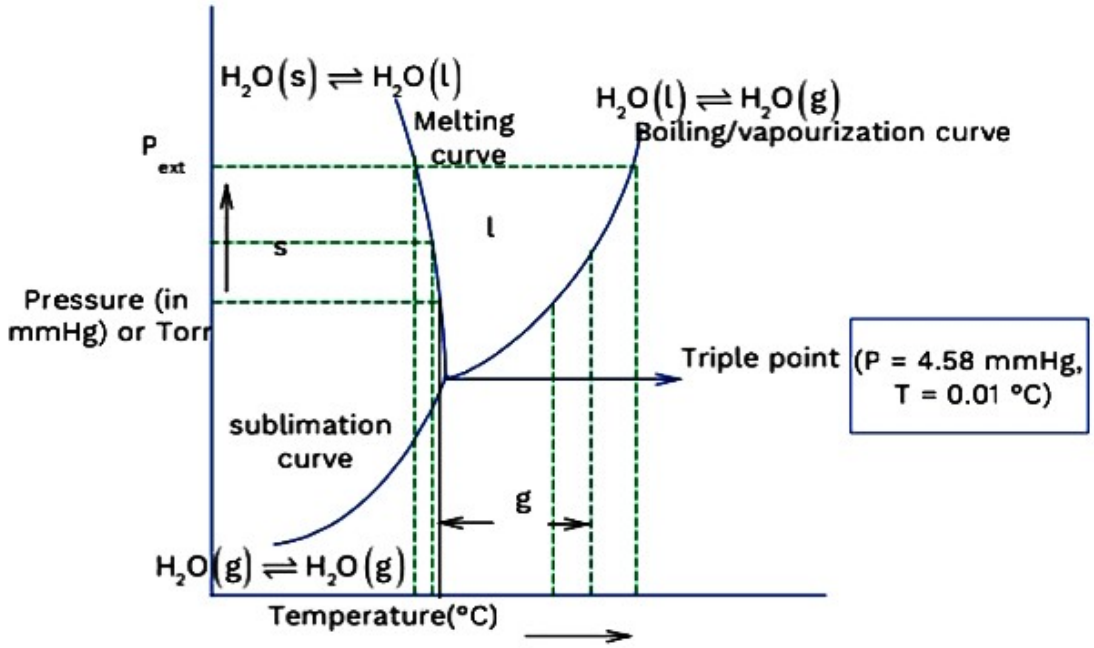
$C(s, \text{graphite}) \rightleftharpoons C(s, \text{diamond})$

Remember ΔH = +ve

Density of diamond > Density of graphite
Favourable condition for above reaction:
High temperature, High pressure ($\because$ volume decreases)

Specific volume of graphite > Specific volume of diamond

Three phase diagram of H_2O



Note: All three phases are in equilibrium at triple point.

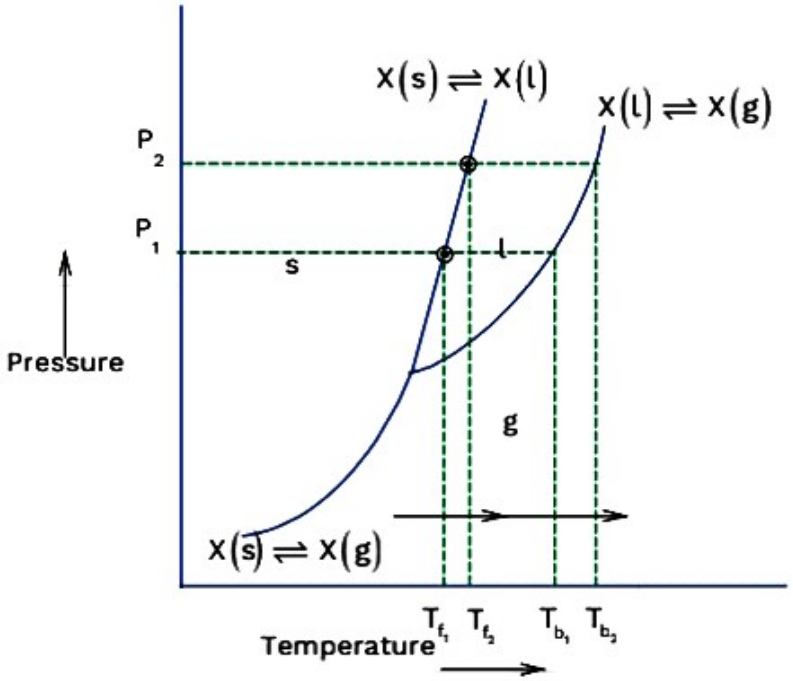
$H_2O(s) \rightleftharpoons H_2O(l)$ $\boxed{\Delta V = -ve}$

Using three phase diagram of H_2O,

If $\boxed{\begin{array}{l} Pr.\uparrow, \text{M.P. of ice} \downarrow \\ Pr.\uparrow, \text{B.P. of ice} \uparrow \end{array}}$

For general Substances:

$X(s) \rightleftharpoons X(l)$ $\Delta V = +ve$



On increasing pressure, melting point increases.
On increasing pressure, boiling point increases.
(as clear from the graph)

Solubility of Solid in Liquid Solvent or Water

Let X(s): solute

$H_2O(l)$: Solvent

1. **If solubility of solid in solvent is endothermic:**

   ΔH = +ve (Endothermic)

   $X(s) + \text{Excess } H_2O(l) \rightleftharpoons X(aq.)$

   If temperature increase, solubility of solid increases following Le Chatelier's Principle.

2. **If solubility of solid in solvent is exothermic:**

   ΔH = −ve (Exothermic)

   $Y(s) + \text{excess } H_2O(l) \rightleftharpoons Y(aq.)$

   If temperature increases, solubility of Y decreases following Le Chatelier's Principle.

   There will be better solubility of Y in cold water.

Solubility of Gas in Liquid or Water:

Like condensation of gas, dissolution of gas in liquid solvent is exothermic process.

$X(g) + \text{excess } H_2O(l) \rightleftharpoons X(aq.)$; ΔH = –ve

For better solubility of gas in liquid or water, suitable conditions are ;

High pressure

Low temperature

e.g. cold drink

